Exact Single-Monomial High-Order Partial Derivatives via Waring Theory and Taylor Directional Derivatives

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Abstract

We present a deterministic Taylor-jet backend for evaluating one fixed high-order mixed partial derivative of a neural-network surrogate, a workload that appears repeatedly in scientific machine learning and high-order physics-informed neural networks. The central observation is that a single multi-index derivative is a coefficient of a directional Taylor polynomial; hence the problem of designing a shortest exact set of directional probes is equivalent to a Waring decomposition of the corresponding monomial. The Carlini–Catalisano–Geramita theorem gives the minimum number of complex directions in closed form and an explicit roots-of-unity construction. We combine this schedule with Taylor-mode propagation through multilayer perceptrons, using the entire activation sinh when complex parameters are used, to obtain an exact floating-point implementation that is differentiable with respect to network parameters. Numerical experiments on direct derivative tasks and a four-dimensional manufactured sixth-order PDE compare the method with nested automatic differentiation and real polarization. The results confirm round-off-level agreement with the direct reference and show that, for repeated-index multi-indices, the Waring schedule reduces the number of Taylor directions while remaining competitive in measured wall-clock time.

Keywords. Physics-informed neural networks; high-order automatic differentiation; Waring decomposition; apolarity; Taylor-mode AD; single-monomial partial derivative.

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1 Introduction

1.1 The problem

Many scientific machine-learning workflows ask, repeatedly during training, for the value of a single fixed high-order mixed partial derivative of a neural-network surrogate. Examples include physics-informed neural networks (PINNs) [11] for high-order PDEs (Korteweg–de Vries, Cahn–Hilliard, biharmonic, polyharmonic), neural-network variational Monte Carlo [10, 5], and stochastic optimal control via the Hamilton–Jacobi–Bellman equation.

The naive evaluation chains successive reverse-mode differentiations through the network. The resulting autograd graph deepens linearly with both the differentiation order and the network depth, and the wall-clock cost becomes the bottleneck of training as soon as the order moves beyond two or three. Three classes of acceleration techniques have been proposed:

(i) **Forward-mode / Taylor-mode automatic differentiation** [1, 4], which propagates truncated Taylor coefficients through every primitive of the network in a single forward pass and avoids the autograd-graph chaining overhead.

(ii) **Stochastic estimators** [6, 12], which write the target operator as an expectation over random tangent vectors and estimate it by Monte Carlo sampling.

(iii) **Operator-specific specializations** [9, 3] for the Laplacian (the trace of the Hessian), exploiting the fact that the trace collapses cleanly through linear layers.

Each has limitations. Forward-mode AD by itself does not say how to combine information from several directions. Stochastic estimators introduce a variance that depends on the sampling budget. And operator-specific methods do not extend to non-trace multi-indices.

This paper develops a fourth, complementary route: an exact deterministic backend for one fixed multi-index at a time, based on identifying the directional schedule problem with a classical combinatorial-algebraic problem – the Waring decomposition of a monomial.

1.2 Contributions

1. **Bridge theorem (Section 3).** We prove that a directional schedule – a finite linear combination of directional Taylor coefficients reproducing the target derivative for every smooth input – exists with a given direction count if and only if a single monomial of degree equal to the differentiation order admits a Waring decomposition of that length. Consequently, the smallest possible direction count equals the Waring rank of the corresponding monomial.

2. **Closed-form schedule (Section 3).** The Carlini–Catalisano–Geramita rank theorem [2] and its roots-of-unity construction yield, for any fixed multi-index, an explicit complex schedule whose direction count attains the information-theoretic minimum over the complex field. For some repeated-index patterns this strictly improves on classical real polarization; for square-free patterns the direction counts coincide.

3. **End-to-end algorithm (Section 4).** We pair the schedule above with a forward-mode Taylor-jet propagation through sinh-activated multilayer perceptrons. Because sinh is entire on the complex plane, the jet rules are well-defined on complex inputs and complex parameters, and the resulting algorithm is exact up to floating-point round-off.

4. **Scientific-computing evaluation (Section 5).** We benchmark direct derivative queries and a four-dimensional manufactured sixth-order PDE. The comparison uses the two implemented deterministic baselines – direct nested automatic differentiation and real polarization with Taylor jets – and reports both direction counts and measured wall-clock costs. Within a fixed wall-clock budget, the complex Waring backend gives the best result in the single tested PDE run; we report this as fixed-budget evidence rather than a statistical superiority claim.

1.3 Scope and limitations

This work treats one fixed multi-index at a time. We do *not* address operator sums such as the Laplacian, polyharmonic operators, or generic Hessian contractions; for these, trace-collapsing techniques [9, 3] are more suitable. We also do not pursue stochastic estimation; its accuracy-cost tradeoffs are complementary to ours [12].

1.4 Relation to existing approaches

The method sits between general automatic differentiation and operator-specific scientific-computing accelerators. Taylor-mode AD [1, 4, 13] supplies the univariate directional derivatives, but it does not determine which finite set of directions extracts a prescribed mixed partial. Real polarization provides a deterministic schedule with real directions, at the cost of up to 2^{p-1} coalesced probes. Stochastic derivative estimators [6, 12] address broader tensor contractions through sampling and are therefore complementary rather than direct replacements for an exact single-monomial schedule. Trace-specialized methods for Laplacians [9, 3] exploit operator sums and are outside the single-monomial setting. Our contribution is to use the monomial Waring rank to make the deterministic direction schedule minimal over $\mathbb{C}$ for the fixed multi-index problem.

1.5 Notation

Throughout the paper, $u : \mathbb{R}^d \to \mathbb{R}$ denotes a sufficiently smooth scalar field. A multi-index of order p is an ordered tuple $\alpha = (i_1, \dots, i_p) \in \{1, \dots, d\}^p$, encoding the operator

$$\partial^\alpha \;:=\; \partial_{i_1}\,\partial_{i_2}\,\cdots\,\partial_{i_p}. \tag{1}$$

The *active coordinates* of α are its distinct entries, relabelled as $i_0, \dots, i_n$ so that their multiplicities $a_0, \dots, a_n$ satisfy $1 \le a_0 \le \cdots \le a_n$ (ties may be broken by the original coordinate index). This ordering is not, in general, the coordinate ordering; it singles out a minimum-multiplicity variable i_0, as required by the monomial Waring rank formula below. We set

$$p \;=\; \sum_{j=0}^{n} a_j, \qquad \alpha! \;=\; \prod_{j=0}^{n} a_j!, \qquad m_j \;=\; a_j + 1. \tag{2}$$

The *active subspace* V is the linear span of the active coordinate directions,

$$V \;:=\; \operatorname{span}\{e_{i_0}, \dots, e_{i_n}\} \;\subset\; \mathbb{R}^d; \tag{3}$$

since ∂^α is insensitive to coordinates outside the active set, all directional probes v below may be taken in V.

The directional Taylor coefficient of order p at x along $v \in V$ is

$$T_p(x; v) \;:=\; \frac{1}{p!}\,\frac{d^p}{d\tau^p}\,u(x + \tau v)\Big|_{\tau=0} \;=\; \frac{1}{p!}\,D^p u(x)[v, \dots, v]. \tag{4}$$

Throughout the paper, the default scalar field is the real field $\mathbb{R}$; we write $\mathbb{R}[z]_p$ for the space of homogeneous polynomials of degree p in $n+1$ variables $z = (z_0, \dots, z_n)$ over $\mathbb{R}$. The complex extension $\mathbb{C}[z]_p$ is needed only in Section 3.2, where the closed-form schedule of Theorem 3.6 uses complex coefficients and complex linear forms; we introduce it explicitly there.

2 Single-Monomial Partials as Polynomial Coefficient Extraction

2.1 The generating polynomial

For $z = (z_0, \dots, z_n) \in \mathbb{R}^{n+1}$ parametrizing the active subspace via $v = \sum_j z_j e_{i_j}$, the *generating polynomial* of u at x is

$$Q_u(z) \;:=\; T_p\Big(x;\, \sum_{j=0}^{n} z_j e_{i_j}\Big), \qquad z \in \mathbb{R}^{n+1}. \tag{5}$$

By construction Q_u is a degree-p homogeneous polynomial in z; the next lemma identifies its coefficients with the partial derivatives of u.

Lemma 2.1 (Generating polynomial expansion)**.** *For $u \in C^p$ on a neighborhood of x,*

$$Q_u(z) \;=\; \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!}\, z^\beta, \tag{6}$$

where β ranges over multi-indices on the active variables and $\partial^\beta = \partial_{i_0}^{\beta_0} \cdots \partial_{i_n}^{\beta_n}$.

Proof. By the symmetric multilinearity of $D^p u(x)$ and the multinomial theorem,

$$D^p u(x)\Big[\sum_j z_j e_{i_j}, \ldots, \sum_j z_j e_{i_j}\Big] \;=\; \sum_{|\beta|=p} \tfrac{p!}{\beta!}\, z^\beta\, \partial^\beta u(x).$$

Dividing by $p!$ yields (6). $\square$

Corollary 2.2. *$\partial^\alpha u(x) = \alpha!\,[z^\alpha]\,Q_u(z)$, where $[z^\alpha]$ extracts the coefficient of the monomial z^α.*

2.2 Directional schedules

A *directional schedule* of length R for ∂^α is a finite collection $(c_r, v_r)_{r=1}^R \subset \mathbb{R} \times V$ such that

$$\partial^\alpha u(x) \;=\; \sum_{r=1}^{R} c_r\, T_p(x; v_r), \qquad \forall\, u \in C^p_x. \tag{7}$$

Because the schedule is required to hold for every $u \in C^p_x$, the p-jet of u at x may be prescribed arbitrarily on the active variables: for any coefficients q_β, the polynomial $\sum_\beta q_\beta \prod_j (y_{i_j} - x_{i_j})^{\beta_j}$ has $\partial^\beta u(x) = \beta!\, q_\beta$. This observation lets us test identities coefficient by coefficient in $\mathbb{R}[z]_p$.

The central design question of this paper is then: *What is the minimum length R^* of a directional schedule* (7)*, and how is one constructed?* Section 3 reduces this to a classical algebraic question.

3 Equivalence to Monomial Waring Decomposition

3.1 The bridge theorem

For $v \in V$, write $v = \sum_{j=0}^n v_j e_{i_j}$ in active coordinates. We pair this direction with the polynomial variable $z = (z_0, \ldots, z_n)$ by $v\cdot z = \sum_{j=0}^n v_j z_j$.

Theorem 3.1 (Bridge)**.** *Let $R \in \mathbb{N}$, $c_r \in \mathbb{R}$, $v_r \in V$, $r = 1, \ldots, R$. The following are equivalent.*

(1) *(Directional schedule)* $\partial^\alpha u(x) = \sum_{r=1}^R c_r T_p(x; v_r) \quad \forall\, u \in C^p_x$.

(2) *(Polynomial identity in $\mathbb{R}[z]_p$)* $p!\, z^\alpha = \sum_{r=1}^R c_r (v_r\cdot z)^p$.

Proof. Write $v_r = (v_{r,0}, \ldots, v_{r,n})$ in active coordinates. By Lemma 2.1,

$$T_p(x; v_r) = \sum_{|\beta|=p} \frac{\partial^\beta u(x)}{\beta!}\, v_r^\beta.$$

Substituting this expansion into statement (1), and using that the order-p active jet of u at x can be prescribed arbitrarily, shows that (1) is equivalent to the finite-dimensional coefficient system

$$\sum_{r=1}^{R} c_r\, v_r^\beta \;=\; \alpha!\, \delta_{\alpha\beta}, \qquad \forall\, \beta \in \mathbb{Z}^{n+1}_{\geq 0} \text{ with } |\beta| = p. \tag{$\star$}$$

On the other hand, expanding statement (2) gives $(v_r\cdot z)^p = \sum_{|\beta|=p} \binom{p}{\beta} v_r^\beta z^\beta$ and hence, after comparing coefficients of z^β,

$$\binom{p}{\beta} \sum_{r=1}^{R} c_r\, v_r^\beta \;=\; p!\, \delta_{\alpha\beta}, \quad \forall\, |\beta| = p.$$

Using $p!/\binom{p}{\beta} = \beta!$ and $\beta!\, \delta_{\alpha\beta} = \alpha!\, \delta_{\alpha\beta}$, this is exactly ($\star$). Thus (1) and (2) are equivalent. $\square$

Definition 3.2 (Waring rank)**.** For a homogeneous polynomial $F \in \mathbb{R}[z]_p$, its *real Waring rank* is

$$R_{\mathbb{R}}(F) \;:=\; \min\Big\{R:\; F = \sum_{r=1}^{R} c_r L_r(z)^p,\; c_r \in \mathbb{R},\; L_r \in \mathbb{R}[z]_1\Big\}.$$

The *complex Waring rank* $R_{\mathbb{C}}(F)$ is defined identically with $\mathbb{R}$ replaced by $\mathbb{C}$; since $\mathbb{R} \subset \mathbb{C}$, $R_{\mathbb{C}}(F) \le R_{\mathbb{R}}(F)$.

Corollary 3.3. *The minimum length R^* of a directional schedule for ∂^α equals the real Waring rank $R_{\mathbb{R}}(z^\alpha)$ of the monomial.*

Remark 3.4 (Complex extension)**.** Theorem 3.1 and its proof go through verbatim if the coefficients c_r and the directions v_r are allowed to take values in $\mathbb{C}$ and $\mathbb{C}^{n+1}$, and the polynomial identity is read in $\mathbb{C}[z]_p$. The minimum length of such a complex schedule then equals $R_{\mathbb{C}}(z^\alpha)$. The next subsection gives this complex rank in closed form.

3.2 The Waring rank of a monomial

The closed-form rank over $\mathbb{C}$ is classical. (Problem (P)/(Q) is posed over $\mathbb{R}$ by default; passing to $\mathbb{C}$ can only decrease the rank, so $R_{\mathbb{C}} \le R_{\mathbb{R}}$.)

Theorem 3.5 (Carlini–Catalisano–Geramita [2])**.** *For $z^\alpha = z_0^{a_0} \cdots z_n^{a_n}$ with $1 \le a_0 \le \cdots \le a_n$,*

$$R_{\mathbb{C}}(z^\alpha) \;=\; \prod_{j=1}^{n} (a_j + 1) \;=\; \frac{\prod_{j=0}^{n} (a_j + 1)}{a_0 + 1}. \tag{8}$$

The lower bound, sketched in [2], uses the apolar ideal $I_{z^\alpha} = (\partial_0^{a_0+1}, \ldots, \partial_n^{a_n+1})$ (a complete intersection); we recall the argument in Appendix A.3. The upper bound is constructive.

Theorem 3.6 (Roots-of-unity directional schedule)**.** *Let $m_j = a_j + 1$ for $j = 1, \ldots, n$, and for each tuple $\zeta = (\zeta_1, \ldots, \zeta_n)$ with $\zeta_j^{m_j} = 1$ define*

$$v_\zeta = e_{i_0} + \sum_{j=1}^{n} \zeta_j e_{i_j}, \qquad c_\zeta = \frac{\alpha!}{\prod_{j=1}^{n} m_j} \prod_{j=1}^{n} \zeta_j. \tag{9}$$

Then $\partial^\alpha u(x) = \sum_\zeta c_\zeta\, T_p(x; v_\zeta)$ for every $u \in C_x^p$. The schedule has length $\prod_{j=1}^{n} m_j = R_{\mathbb{C}}(z^\alpha)$ and is therefore length-optimal over $\mathbb{C}$.

The proof, given in Appendix A.2, verifies condition (2) of Theorem 3.1 directly via the roots-of-unity filter $\sum_{\zeta^m=1} \zeta^k = m\, \mathbf{1}_{m|k}$.

To illustrate how Theorem 3.1 couples the analytic side (7) (problem (P)) with the algebraic side $p!\, z^\alpha = \sum_r c_r (v_r \cdot z)^p$ (problem (Q)), we work out two examples: a low-order one for which both sides reduce to hand-checkable identities, and a higher-order repeated-index one that exhibits the full strength of the complex Waring construction.

Example 3.7 ($p = 2$, $\alpha = (1, 1)$, real schedule of length 2)**.** For $\alpha = (1, 1)$ we have $n = 1$, $a_0 = a_1 = 1$, $m_1 = 2$, $U_2 = \{+1, -1\}$. Equation (9) produces the schedule

$$\big(c_{+1}, v_{+1}\big) = \big(\tfrac{1}{2},\, e_1 + e_2\big), \qquad \big(c_{-1}, v_{-1}\big) = \big(-\tfrac{1}{2},\, e_1 - e_2\big). \tag{10}$$

Verification on the analytic side. Expanding the directional Taylor coefficient $T_p(x; v) = \frac{1}{2}\big(u_{11} v_1^2 + 2 u_{12} v_1 v_2 + u_{22} v_2^2\big)$,

$$\tfrac{1}{2} T_p(x; e_1 + e_2) - \tfrac{1}{2} T_p(x; e_1 - e_2) = \tfrac{1}{4}\big[(u_{11} + 2u_{12} + u_{22}) - (u_{11} - 2u_{12} + u_{22})\big] = u_{12}(x).$$

Verification on the algebraic side. The equivalent polynomial identity in $\mathbb{R}[z]_2$ reads

$$2!\, z_0 z_1 \;=\; \tfrac{1}{2}(z_0 + z_1)^2 - \tfrac{1}{2}(z_0 - z_1)^2 \;=\; \tfrac{1}{2}\big[(z_0^2 + 2z_0 z_1 + z_1^2) - (z_0^2 - 2z_0 z_1 + z_1^2)\big] \;=\; 2\, z_0 z_1,$$

matching $p!\, z^\alpha = 2\, z_0 z_1$. The two checks coincide via the bridge of Theorem 3.1.

Example 3.8 ($p = 6$, $\alpha = (2, 2, 2)$, complex schedule of length 9)**.** Consider $u_{112233} = \partial_1^2\partial_2^2\partial_3^2 u$, i.e. $\alpha = (2, 2, 2)$. Here $n = 2$, $a_0 = a_1 = a_2 = 2$, $m_1 = m_2 = 3$; take $\omega = e^{2\pi i/3}$ and $U_3 = \{1, \omega, \omega^2\}$. Equation (9) produces $R_{\mathbb{C}}(z^{(2,2,2)}) = 9$ directions: for each $(\zeta_1, \zeta_2) \in U_3 \times U_3$,

$$v_\zeta \;=\; e_1 + \zeta_1 e_2 + \zeta_2 e_3, \qquad c_\zeta \;=\; \frac{(2!)^3}{3 \cdot 3}\,\zeta_1\zeta_2 \;=\; \frac{8}{9}\,\zeta_1\zeta_2. \tag{11}$$

Analytic side. The directional schedule reads

$$u_{112233}(x) \;=\; \frac{8}{9}\sum_{\zeta_1,\zeta_2\in U_3} \zeta_1\zeta_2\, T_p\big(x;\, e_1 + \zeta_1 e_2 + \zeta_2 e_3\big). \tag{12}$$

Algebraic side. The equivalent polynomial identity in $\mathbb{C}[z]_6$ is

$$6!\, z_0^2 z_1^2 z_2^2 \;=\; \frac{8}{9}\sum_{\zeta_1,\zeta_2\in U_3} \zeta_1\zeta_2\,(z_0 + \zeta_1 z_1 + \zeta_2 z_2)^6. \tag{13}$$

That (13) holds is verified by the roots-of-unity filter (Lemma A.1): expanding the right-hand side multinomially and collecting the coefficient of $z_0^{\beta_0} z_1^{\beta_1} z_2^{\beta_2}$ reduces to the product $\prod_{j=1}^{2}\sum_{\zeta_j\in U_3}\zeta_j^{1+\beta_j}$, which is non-zero exactly when $\beta_j \equiv 2 \pmod 3$ for $j = 1, 2$; together with $\sum_j \beta_j = 6$ this forces $\beta = (2, 2, 2)$, and the leading constant $\frac{8}{9}\cdot\frac{6!}{2!\,2!\,2!}\cdot 9 = 720 = 6!$ gives the right normalisation.

By comparison, the coalesced antipodally merged real polarization schedule used in our baseline requires 13 distinct nonzero directions for the same $u_{112233}(x)$; the complex Waring schedule above attains the information-theoretic complex lower bound $R_{\mathbb{C}} = 9$ (Theorem 3.5).

3.3 Pattern taxonomy

Table 1 lists the rank values for representative active-exponent patterns up to $p = 6$, comparing the complex Waring rank $R_{\mathbb{C}}$ of Theorem 3.5 with the post-antipodal polarization count 2^{p-1}, the latter being the direction count attainable by classical real polarization for a generic multi-index of order p.

pattern	example multi-index	$R_{\mathbb{C}}$	polarization
(p)	$\partial_1^p u$	1	$\le 2^{p-1}$
$(2,1)$	u_{112}	3	3
$(1,1,1)$	u_{123}	4	4
$(3,1)$	u_{1112}	4	4
$(2,2)$	u_{1122}	**3**	4
$(2,1,1)$	u_{1123}	6	6
$(1,1,1,1)$	u_{1234}	8	8
$(4,2)$	u_{111122}	**5**	7
$(2,2,2)$	u_{112233}	**9**	13
(1^6)	u_{123456}	32	32

Table 1: Bold entries mark patterns where the complex Waring schedule is strictly shorter than the polarization count. The square-free pattern (1^p) saturates the lower bound 2^{p-1}; the method of this paper offers no improvement there.

4 Taylor-Mode Evaluation and Backend Dispatch

4.1 The Taylor jet for Linear–sinh MLPs

Given a single direction v, Theorem 3.6 reduces the problem to evaluating $T_p(x; v)$. We use forward-mode Taylor propagation [1, 4], replacing each intermediate activation

$y \in \mathbb{R}^B$ by a length-$(p+1)$ tuple $(y_0, \dots, y_p)$ representing its truncated Taylor coefficients $y_k = (1/k!)\, d^k y/d\tau^k|_{\tau=0}$.

For an MLP $u = \phi_L \circ A_L \circ \cdots \circ \phi_1 \circ A_1$ with affine A_ℓ and entrywise activations ϕ_ℓ, the jet rules are as follows. The affine layer $y = Wx + b$ propagates each jet order independently:

$$y_0 \;=\; Wx_0 + b, \qquad y_k \;=\; Wx_k \quad (k \geq 1). \tag{14}$$

The tanh activation, with auxiliary jet $z = 1 - y^2$, satisfies

$$y_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, z_j\, x_{k-j}, \qquad z_k \;=\; -\sum_{j=0}^{k} y_j\, y_{k-j}, \qquad k \geq 1. \tag{15}$$

The sinh activation, with auxiliary jet $z = \cosh(x)$, satisfies the coupled recurrence

$$y_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, z_j\, x_{k-j}, \qquad z_k \;=\; \frac{1}{k}\sum_{j=0}^{k-1}(k-j)\, y_j\, x_{k-j}, \qquad k \geq 1. \tag{16}$$

The sinh recurrence is identical to that of sin / cos but without the sign flip. These recurrences are algebraic identities and can be evaluated on complex tensors. For the global complex-direction backend used in the PDE experiment we use sinh, because sinh and cosh are entire on $\mathbb{C}$. The tanh rule is kept for real-valued experiments and local complex evaluations, but tanh is meromorphic and has poles in the complex plane.

Each Taylor coefficient y_k is a regular tensor, and the jet propagation has no autograd-graph chaining, so storage is $(p+1)$ times the ordinary forward-pass activation size. We additionally use a custom `torch.autograd.Function` for each activation jet, with the closed-form backward

$$g_x^{(m)} \;=\; \sum_{j=0}^{p-m} \overline{q_j}\, g_y^{(m+j)}, \qquad q \;=\; \phi', \tag{17}$$

which avoids reverse differentiation through the intra-jet recurrence and is essential for memory-efficient training.

4.2 Backends

We compare three implemented deterministic backends for evaluating one fixed $\partial^\alpha u(x)$. All three are differentiable in the network parameters θ, so each can be used directly as the residual term of a PINN loss. The implementation also provides an `auto` dispatch option, but it is a selection rule between B2 and B3 rather than a separate mathematical backend.

(B1) Direct nested AD. The reference baseline performs p successive calls to PyTorch's `autograd.grad`, each with retained computation graph. This induces a reverse-mode autograd graph of depth $O(pL)$ for a depth-L network.

(B2) Real polarization with Taylor jet. The classical antipodally merged polarization identity is

$$\partial^\alpha u(x) \;=\; \frac{1}{2^p}\sum_{\epsilon \in \{\pm 1\}^p} \Big(\prod_{r=1}^p \epsilon_r\Big)\, T_p\Big(x;\, \sum_{r=1}^p \epsilon_r\, e_{i_r}\Big). \tag{18}$$

After antipodal merging this yields at most 2^{p-1} real directions. The directional Taylor coefficients $T_p(x;\,\cdot\,)$ are evaluated by the jet propagation of Section 4.

(B3) Complex Waring with Taylor jet. The roots-of-unity construction of Theorem 3.6 produces $R_{\mathbb{C}}(z^\alpha) = \prod_{j=1}^n (a_j + 1)$ directions in $\mathbb{C}^{n+1}$; for several repeated-index patterns this is smaller than the coalesced polarization count of (B2). For real-parameter networks, the complex directions induce per-call casts of the linear weights via the differentiable `Tensor.to` operator; for complex-parameter networks (Section 5.4) no cast is required and the network itself acts on the complex directions natively.

4.3 The end-to-end algorithm

Algorithm 1 summarises the complete pipeline of backend (B3): given a smooth model u (an MLP whose primitives are covered by the jet rules (14)–(16)), a point x, and a multi-index α of order p, it returns the value $\partial^\alpha u(x)$ without any nested back-propagation. Backend (B2) follows the same overall structure with the Waring schedule (lines 3–7) replaced by the polarization schedule (18).

Algorithm 1 Single-monomial $\partial^\alpha u(x)$ via complex Waring schedule + Taylor jet.

Require: smooth scalar field u; point $x \in \mathbb{R}^d$; multi-index α of order p

Ensure: the value $\partial^\alpha u(x)$

1: relabel the active coordinates of α as $i_0, \dots, i_n$ with multiplicities $1 \le a_0 \le \cdots \le a_n$
2: $m_j \leftarrow a_j + 1$ for $j = 1, \dots, n$; set $\mathcal{I} \leftarrow U_{m_1} \times \cdots \times U_{m_n}$
3: initialise schedule $\mathcal{S} \leftarrow \emptyset$
4: **for all** $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathcal{I}$ **do**
5: $\quad v_\zeta \leftarrow e_{i_0} + \sum_{j=1}^n \zeta_j \, e_{i_j}$
6: $\quad c_\zeta \leftarrow \dfrac{\alpha!}{\prod_{j=1}^n m_j} \prod_{j=1}^n \zeta_j$
7: $\quad \mathcal{S} \leftarrow \mathcal{S} \cup \{(c_\zeta, v_\zeta)\}$
8: **end for**
9: result $\leftarrow 0$
10: **for all** $(c, v) \in \mathcal{S}$ **do**
11: $\quad$ form the input jet $\mathbf{J}_0 \leftarrow (x, v, 0, \dots, 0)$ of length $p+1$
12: $\quad \mathbf{J}_L \leftarrow$ jet propagation of $\mathbf{J}_0$ through every primitive of u using (14)–(16)
13: $\quad T_p(x; v) \leftarrow [\mathbf{J}_L]_p$ $\quad \triangleright$ order-p coefficient of the output jet
14: $\quad$ result $\leftarrow$ result $+ \, c \cdot T_p(x; v)$
15: **end for**
16: **return** result

Remark 4.1. By Theorem 3.1 together with Theorem 3.6, Algorithm 1 returns $\partial^\alpha u(x)$ exactly (up to floating-point round-off in the jet propagation). The number of inner iterations is $|\mathcal{S}| = R_{\mathbb{C}}(z^\alpha)$, which is information-theoretically optimal over $\mathbb{C}$ (Theorem 3.5).

4.4 Cost and numerical considerations

For a fixed direction, the Taylor jet stores $p+1$ coefficient tensors per activation and performs the same linear layers at each jet order; there is no chain of nested reverse-mode differentiation through the input coordinates. The total value-evaluation cost of B3 is therefore proportional to $R_{\mathbb{C}}(z^\alpha)$ directional jet evaluations, while B2 scales with the coalesced polarization count R_{pol}. The rank reduction is an algebraic cost reduction; the realized wall-clock time also depends on complex arithmetic, batching, activation kernels, and whether parameter gradients are required.

The schedule coefficients are roots of unity multiplied by $\alpha!/R_{\mathbb{C}}$, so coefficient magnitudes are explicit and do not come from solving an ill-conditioned linear system. The method has no finite-difference step size and no Monte Carlo variance. Remaining errors are ordinary floating-point round-off from Taylor recurrences, complex arithmetic, and the final summation over directions.

5 Numerical Experiments

We evaluate three claims:

(C1) The Waring + Taylor-jet backend matches direct nested AD to round-off accuracy on the value of $\partial^\alpha u(x)$, and supports parameter-gradient back-propagation in PINN training.

(C2) For repeated-index patterns where $R_\mathbb{C}$ is strictly smaller than the coalesced polarization count, the complex-Waring backend uses fewer Taylor directions; measured wall-clock differences are reported rather than inferred solely from rank.

(C3) On a 4D manufactured sixth-order PDE, training a complex-parameter sinh PINN with the Waring backend has a much smaller time-per-step than nested AD and reaches the lowest error in the single-seed fixed-budget run reported here.

5.1 Experimental setup

Hardware. Single-GPU NVIDIA T4 (16 GB), CUDA 12.1, PyTorch 2.5.

Networks. The direct-derivative microbenchmark uses a real Linear–sinh MLP with hidden width 64 and depth 4, matching the saved run in `results/quick_compare_T4_20260602_0303.txt`. The PDE experiment uses a complex-parameter Linear–sinh MLP with hidden width 32 and depth 4. In the PDE loss, the physical solution is $\Re u$; imaginary parts of the weights are regularized via a Tikhonov term $\lambda \|\operatorname{Im} W\|^2$ with $\lambda = 10^{-6}$.

Precision and reference. The microbenchmark runs in fp64, while the PDE training uses complex128 parameters and activations. Reference values for derivative accuracy are computed with backend B1 (direct nested AD) on the same network instance.

Timing protocol. The microbenchmark reports median wall-clock per derivative call over 60 timed repetitions after 5 warm-up calls, as recorded in the saved result file. The PDE experiment gives each backend a 60-second wall-clock training budget and reports the realized per-step time and peak allocated GPU memory. The saved PDE run uses one random seed shared across backends; it is intended as a controlled fixed-budget comparison, not as an uncertainty-quantified training study.

5.2 Direct derivative microbenchmark

We sample $B = 8$ inputs in $\mathbb{R}^d$ for $d = 8$ and measure the cost of computing one $\partial^\alpha u$ for each pattern in Table 1. Table 2 reports the wall-clock per call for the three backends on selected patterns.

pattern	p	$R_\mathbb{C}$	R_{pol}	B1 (ms)	B2 (ms)	B3 (ms)
(3)	3	1	2	3.399	2.399	2.532
(4)	4	1	2	7.679	3.302	3.425
$(2,2)$	4	**3**	4	7.689	3.314	3.458
(6)	6	1	3	52.253	5.916	5.837
$(4,2)$	6	**5**	7	52.373	5.918	5.904
$(2,2,2)$	6	**9**	13	52.219	5.911	5.916
(1^6)	6	32	32	52.323	5.961	5.943

Table 2: Median wall-clock per single $\partial^\alpha u$ call on the T4, with $d = 8$, $B = 8$, hidden width 64, depth 4, fp64, and a real-parameter sinh MLP. Times are from 60 timed repetitions after 5 warm-up calls. Bold entries mark patterns where $R_\mathbb{C} < R_{\text{pol}}$. The largest reported relative error of a jet backend against B1 is below $6 \cdot 10^{-15}$.

The relative error of B2 and B3 against B1 is at the round-off level. At $p = 6$ the jet path is roughly 8.8× faster than nested AD in this value-only benchmark. This verifies the accuracy part of C1 for the patterns tested; parameter-gradient support is exercised in the PINN experiment below.

5.3 Repeated-index regime

For repeated patterns, the algebraic advantage of B3 is a smaller number of Taylor directions: for example, $(4, 2)$ uses 5 directions instead of 7, and $(2, 2, 2)$ uses 9 instead of 13. Table 2 shows that this direction-count reduction does not automatically translate into a large value-only wall-clock gain on the tested T4 configuration: B2 and B3 are within a few percent, with the complex arithmetic overhead partly offsetting the shorter schedule. We therefore treat C2 as a complexity and direction-count advantage, and report measured timings case by case rather than claiming a universal wall-clock ordering.

5.4 PINN experiment: 4D manufactured sixth-order PDE

Problem. Consider the manufactured PDE on $\Omega = (-1, 1)^4$:

$$\partial_{x_1}^4 \partial_{x_2}^2 u(x) \;=\; f(x), \qquad u|_{\partial\Omega} \text{ given}, \tag{19}$$

with $u_{\text{exact}}(x) = \sinh(x_1)\cos(x_2)\,\mathrm{e}^{-(x_3^2+x_4^2)/4}$. Since $\sinh^{(4)} = \sinh$ and $\cos^{(2)} = -\cos$, the source has the closed form $f = -u_{\text{exact}}$. The dominant operator is the single multi-index with active-exponent pattern $(4, 2)$; by Theorem 3.5, $R_{\mathbb{C}} = 5$ versus polarization count 7.

Architecture and training. We use the complex network specified in the experimental setup (hidden width 32, depth 4) with $d = 4$, input embedded as $x \mapsto x + 0\,\mathrm{i}$. The PINN loss combines an interior residual term, a boundary-fit term, and a Tikhonov regularizer on the imaginary part of the weights:

$$\mathcal{L} \;=\; \mathcal{L}_{\text{int}} \;+\; \beta\,\mathcal{L}_{\text{bc}} \;+\; \lambda \sum_{W} \|\Im W\|^2, \tag{20}$$

with

$$\mathcal{L}_{\text{int}} \;=\; \mathbb{E}_{x\sim U(\Omega)}\big|\Re\partial^{(4,2)}u(x) - f(x)\big|^2, \qquad \mathcal{L}_{\text{bc}} \;=\; \mathbb{E}_{x\sim U(\partial\Omega)}\big|\Re u(x) - u_{\text{exact}}(x)\big|^2,$$

$\beta = 10^2$ and $\lambda = 10^{-6}$. Training uses Adam with learning rate 10^{-3}; each step samples $N_{\text{int}} = 128$ interior points and $N_{\text{bc}} = 64$ boundary points.

Comparison protocol. Each backend is given the same wall-clock budget of 60 seconds. We report *steps completed within the budget*, the wall-clock per step, peak GPU memory, the final value of $\mathcal{L}_{\text{int}}$ (averaged over the last 20 steps), and the relative L^2 error of $\Re u$ against u_{exact} on a held-out grid of size 4096. Results in Table 3 are measured directly on the T4 with the implementation of this paper. For a submission-quality accuracy comparison between B2 and B3, the same protocol should be repeated over multiple seeds with a fixed and recorded held-out evaluation set.

backend	steps	ms/step	peak (MB)	final $\mathcal{L}_{\text{int}}$	rel. L^2 error
B3 (complex Waring + jet)	4248	**14.1**	**40**	$\mathbf{1.74\times10^{-3}}$	$\mathbf{8.50\times10^{-3}}$
B2 (polarization + jet)	4192	14.3	49	6.32×10^{-3}	4.43×10^{-2}
B1 (direct nested AD)	211	285.6	103	2.82×10^{-1}	2.49×10^{-1}

Table 3: PINN training cost and accuracy on the 4D manufactured sixth-order problem (19), identical 60-second wall-clock budget per backend. Same network, optimizer, sampling rule and seed are used across rows. This is a single-seed run: the complex Waring backend reaches the lowest interior loss and the smallest L^2 error here, while nested AD completes only 211 steps and remains far from convergence.

Three observations are in order. First, the two jet-based backends share essentially identical per-step cost (14.1 and 14.3 ms), which is roughly $20\times$ faster than nested AD; nested AD's peak memory is 2.5× higher because of the deeper autograd graph. Second, B3 uses $R_{\mathbb{C}} = 5$ directions, two fewer than polarization ($R_{\mathrm{pol}} = 7$). In this fixed-budget run it also reaches a smaller final L^2 error (8.5×10^{-3} vs. 4.4×10^{-2}), although this accuracy comparison is affected by stochastic sampling during training and should be read together with additional seeds in future work. Third, sinh is entire on $\mathbb{C}$, so complex Waring directions and complex-parameter sinh MLPs compose without any runtime dtype conversion in the inner loop. These results support C3 for this single-seed manufactured problem.

6 Discussion and Limitations

Strengths. Complex Waring is at its strongest when the active-exponent pattern is repeated and balanced, e.g., $(2,2)$, $(2,2,2)$, $(3,3)$, $(4,2)$. For these the rank $\prod_{j\geq1}(a_j+1)$ is much smaller than 2^{p-1}; concretely, $(2,2,2)$ yields rank 9 while polarization needs 13 directions for the same multi-index. PINNs whose residual contains such a single dominant operator – not uncommon in fourth- and sixth-order beam, plate, and Cahn–Hilliard models – benefit directly.

Limitations. Square-free patterns saturate the 2^{p-1} bound over both $\mathbb{R}$ and $\mathbb{C}$; for $\partial_x\partial_y\partial_z\partial_w\partial_u\partial_v$ the method offers no rank advantage over polarization. Operators that are *sums* of multi-indices (Laplacian, biharmonic) fall outside our scope and are best served by trace-collapsing methods [9, 3].

Experimental evidence. The direct-derivative benchmark is a value-mode accuracy and timing test; it verifies round-off agreement and exposes the direction-count advantage, but it does not by itself prove a universal wall-clock ordering between complex Waring and real polarization. The PINN experiment is a single-seed fixed-budget run. It demonstrates that the jet backends reduce per-step cost substantially relative to nested AD on the manufactured problem, while the observed accuracy gap between B3 and B2 should be validated with multiple seeds, fixed evaluation grids, and structured CSV/JSON result files before being used as a statistical comparison.

Connection to STDE. The Stochastic Taylor Derivative Estimator [12] solves a strictly broader problem – arbitrary contraction of the order-k derivative tensor – via stochastic sparse jets driven by integer partitions. For one fixed multi-index α our deterministic Waring schedule is exact, while STDE is an unbiased estimator with variance dictated by its sample budget. The two are complementary: STDE is the natural choice when the operator is a linear combination of many multi-indices; Waring is the natural choice when attention is on one dominant multi-index that one wishes to evaluate exactly.

7 Conclusion

We have shown that the construction of a directional schedule for one mixed partial derivative $\partial^\alpha u$ of order p is exactly the problem of a Waring decomposition of the monomial z^α. The classical Carlini–Catalisano–Geramita rank theorem and its roots-of-unity construction yield an explicit, rank-optimal directional schedule over $\mathbb{C}$; paired with Taylor-mode propagation through sinh-activated MLPs, this gives a deterministic, exact, parameter-differentiable backend for high-order single-monomial residuals. In the tested 4D manufactured sixth-order PDE, the jet backends are much faster per step than nested automatic differentiation, and the complex Waring backend achieves the lowest error in the reported single-seed fixed-budget run.

The method is at its most useful for repeated-index multi-indices, where the complex Waring rank is materially smaller than the coalesced real polarization count. For square-free patterns it offers no rank advantage; for operator sums it is out of scope. Within its niche – a single, dominating, repeated-index high-order operator – it provides an exact deterministic alternative to stochastic estimators, while leaving broader operator contractions to methods designed for that setting.

A Proofs of Directional Schedule Identities

A.1 Roots-of-unity filter

Lemma A.1. *For any positive integer m and integer k, $\sum_{\zeta:\,\zeta^m=1}\zeta^k = m\,\mathbf{1}_{\{m|k\}}$.*

Proof. If $m \mid k$ every term is 1; otherwise the sum is a non-trivial geometric series in the primitive m-th root of unity, which sums to zero. □

A.2 Proof of Theorem 3.6

We verify condition (2) of Theorem 3.1: namely that the schedule (9) satisfies $p!\,z^\alpha = \sum_\zeta c_\zeta (v_\zeta \cdot z)^p$ in $\mathbb{C}[z]_p$.

Expanding $(v_\zeta \cdot z)^p$ by the multinomial theorem and collecting the coefficient of z^β (with $|\beta| = p$):

$$[z^\beta]\sum_\zeta c_\zeta (v_\zeta \cdot z)^p = \frac{\alpha!}{\prod_{j=1}^n m_j}\binom{p}{\beta}\prod_{j=1}^n\Big(\sum_{\zeta_j^{m_j}=1}\zeta_j^{1+\beta_j}\Big).$$

By Lemma A.1, each inner sum equals m_j iff $m_j \mid 1+\beta_j$, i.e. $\beta_j \equiv a_j \pmod{m_j}$, and equals zero otherwise.

A short counting argument shows that the unique feasible joint assignment is $\beta = \alpha$. Indeed, the congruence forces $\beta_j = a_j + t_j(a_j+1)$ for $j \geq 1$ with $t_j \geq 0$. If some $t_j > 0$, then

$$\beta_0 = a_0 - \sum_{j=1}^n t_j(a_j+1) < 0,$$

because $a_j \geq a_0$, impossible. Hence all $t_j = 0$ and $\beta = \alpha$. With $\beta = \alpha$ all the m_j factors cancel and the leading constant becomes $\alpha!\binom{p}{\alpha} = p!$, giving $[z^\alpha]\sum_\zeta c_\zeta(v_\zeta \cdot z)^p = p!$ and all other coefficients vanish. □

A.3 Apolar pairing alternative

The Waring rank $R_{\mathbb{C}}(F)$ of a homogeneous polynomial $F \in \mathbb{C}[z]_p$ admits the apolar characterization $R_{\mathbb{C}}(F) = \min\{\deg X : X \subset \mathbb{P}^n$ a finite reduced scheme apolar to $F\}$, where X is apolar to F iff $I_X \subset F^\perp$ and $F^\perp$ is the apolar (Macaulay inverse) ideal of F. For $F = z^\alpha$, $F^\perp = (\partial_0^{a_0+1}, \ldots, \partial_n^{a_n+1})$, which is a complete intersection. The Hilbert function of the quotient $\mathbb{C}[\partial]/F^\perp$ gives the lower bound matching (8); details in [7, 8].

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